Supplementary Material for the Article: Time–Frequency Scattering for Playing Technique Recognition

Changhong Wang, *Student Member, IEEE,* Emmanouil Benetos, *Senior Member, IEEE,* Vincent Lostanlen, and Elaine Chew

This document provides further details regarding datasets and detection results for our work "Time–Frequency Scattering for Playing Technique Recognition" [1].

A. Existing Computational Research on Playing Techniques

Besides the summary of existing computational research on playing techniques in our paper [1], we provide a complete list of playing techniques analysed and the corresponding methodologies applied in Table I.

B. Additional Information on the Datasets

Table II displays the number of ground truth frames, total duration of the ground truth, and duration of co-articulations for each playing technique in the CBFdataset. As can be seen, the number of samples for each type of playing technique is highly imbalanced. Portamento has 1010 samples while the number of trills is only 197. As compared to the duration of the whole dataset of 157 minutes the co-articulations amount to only 5 minutes. To have comparable results to the extended datasets discussed in our paper [1] and to obtain a confusion matrix between different CBF techniques, we discard the co-articulation samples and conduct a multiclass classification.

Fig. 1 shows the comparison between glissando played in isolation and those performed in full-length pieces. The isolated one consists of consecutive note changes with approximately equal duration, while the performed ones vary from each other in different ways. Player 1 lengthened the first note for the glissando at position 1, while emphasised the beginning of the glissando at position 2 by increasing the intensity. Player 2 performed the glissando together with flutter-tongue. These variabilities requires data collection of playing techniques in music context.

C. Additional Result Analysis

In Section VII-D of our paper [1], we detect all seven CBF playing techniques via a multiclass classifier using the concatenation of the proposed adaptive time–frequency scattering (AdaTFS) and the direction-invariant joint time–frequency scattering (d-jTFS) representations as input. This is based on a performance comparison of the system using the AdaTFS only, the d-jTFS only, and the concatenation of these two features. The concatenated feature achieves the highest marco averaged F-measure of 74.9%, as compared to that of 53.2% using the AdaTFS only and that of 71.7% using the d-jTFS only. Fig. 3 shows the F-measures obtained in the multiclass classification of each playing technique using the three type of scattering representations. Although the d-jTFS obtains comparable result as that of the concatenated feature, the latter

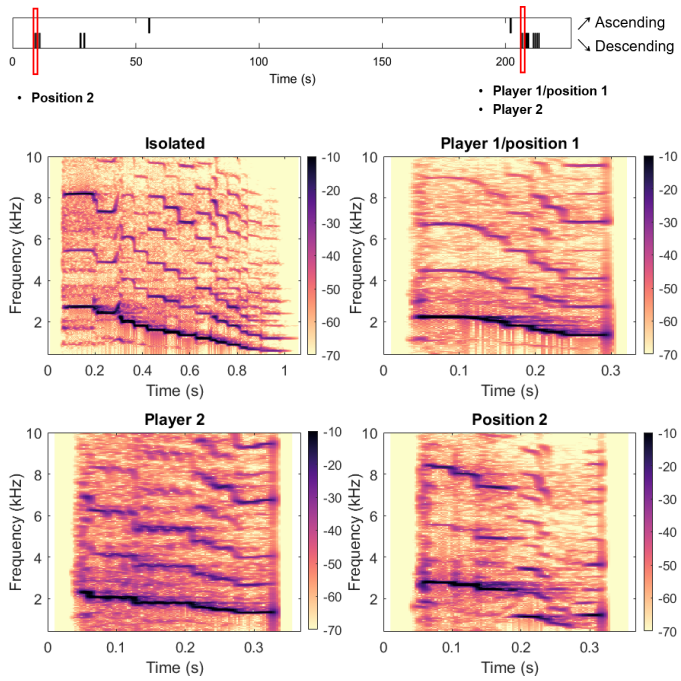

Fig. 1. Variations of glissandi between different players and different temporal positions in the performance of *Busy Delivery Harvest*.

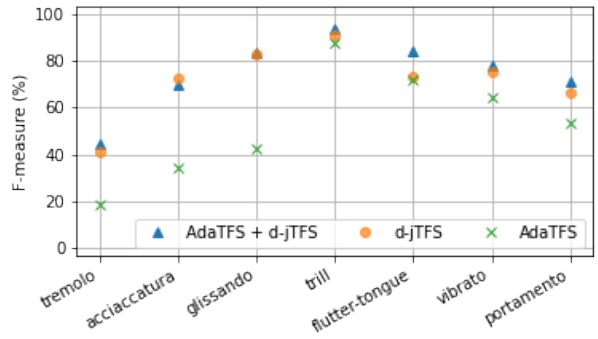

Fig. 2. Performance comparison of the recognition system on the CBFdataset using the proposed AdaTFS, the d-jTFS, and the concatenation of AdaTFS and d-jTFS as input.

TABLE I
SUMMARY OF EXISTING COMPUTATIONAL RESEARCH ON PLAYING TECHNIQUES

Instrument	Expressive patterns	Methodologies applied	Cite
Guitar	Ornamentation defined by rules	Dynamic time warping sequence matching	[2]
	Legato, glissando	Symbolic, aggregate approximation	[3]
	Muted, harmonic, bend, slide, hammer-on, pull-off	Rule-based detection	[4]
	Muting, vibrato, pull-off, hammer on, sliding, bending	Sparse cepstral and phase codes	[5]–[7]
	Plucking: finger, picked, muted, slap; expression: harmonics, vibrato, bending, slide, etc.	Feature extraction, support vector machines with radial basis function kernel	[8], [9]
Piano	Trills	Independent component analysis	[10]
	Pdealling techniques	Supporter vector machines	[11]
Violin	Beginners' faults on playing techniques	K-nearest neighbours	[12]
	Flageolet, non-vibrato, pizzicato, sordino, spiccato, sul ponticello, sul tasto, tremolo	Sparse modelling	[13]
Drums	strike, buzz roll, drag, flam	Non-negative matrix factorization-based activation	[14]
	A variety of drum playing techniques	K-nearest neighbours, partial decision tree, etc.	[15]
Cello	Vibrato, tremolo, trill, glissando, etc.	Folded convolutional neural network	[16]
Irish flute	Cut, strike, crann, roll, shake	Rule-based method; hidden Markov models (HMMs)	[17]
Highland pipe	60 ornamentations	Dynamic time warping	[18]
Ney	Vibrato, kaydirma	Rule-based characterisation	[19]
Erhu	Vibrato, portamento	Filter diagonalisation method, HMMs	[20]
Guqin	None, vibrato, upward/downward/inverted-U/U protamento	Neural network	[21]
Chinese bamboo flute	Vibrato, tremolo, till, flutter-tongue, acciaccatura, portamento, glissando	HMMs, adaptive/joint time–frequency scattering	[22]–[24]
	Trill, flutter-tongue, tremolo	Synthesising	[25], [26]
Other	Vibrato, glissando	Rule-based methods	[27]
Multi-instruments	140 instrumental playing techniques (isolated)	Scattering transform	[28]
Vocal techniques	Vibrato, glissando, linear note change	COSFIRE filter	[29]
	17 different vocal techniques	VocalSet	[30]

TABLE II
NUMBER OF SAMPLES, DURATION OF GROUND TRUTH, AND DURATION OF CO-ARTICULATIONS FOR THE SEVEN TECHNIQUES IN THE CBFDATASET. TOTAL DURATION OF THE CBFDATASET IS 157 MINUTES, INCLUDING 5-MINUTE CO-ARTICULATIONS

Techniques	Flutter-tongue	Trill	Vibrato	Tremolo	Acciaccatura	Portamento	Glissando
Number of samples	628	197	519	555	441	1010	490
Duration of ground truth (minute)	7.9	15.0	13.8	6.0	1.7	6.3	5.0
Duration of co-articulations (minute)	1.3	0.1	3.7	3.7	0	0.9	0.3

outperforms the former for 5 out of the 7 playing techniques.

Fig. 3 and Fig. 4 are the confusion matrices obtained for recognising playing techniques in the SOL dataset and VocalSet. The recognition for the SOL dataset presents few confusions between techniques. This may be attributed to the dataset structured with short clips of individual techniques. No technique is based on the same note of the same instrument, which offers high intra-class variability for playing techniques. For the VocalSet, as can bee seen from the normalised confusion matrix in Fig. 4 (b), the lip trill obtains the best score while the inhaled technique obtains the lowest. There exists a large amount of confusion between breathy and vocal fry. The straight technique exhibits high confusion with other techniques.

Fig. 5 and Fig. 6 are the confusion matrices obtained for playing technique recognition in the CBFdataset using mel-frequency cepstral coefficients (MFCCs) and modulation power spectrum (MPS) as input features to the classifier, respectively. Both features achieve the best score for trill recognition. The MFCCs detect none of the tremolos while the MPS returns the lowest score for acciaccatura recognition. The confusion between vibrato and tremolo is also found for the recognition result by MPS.

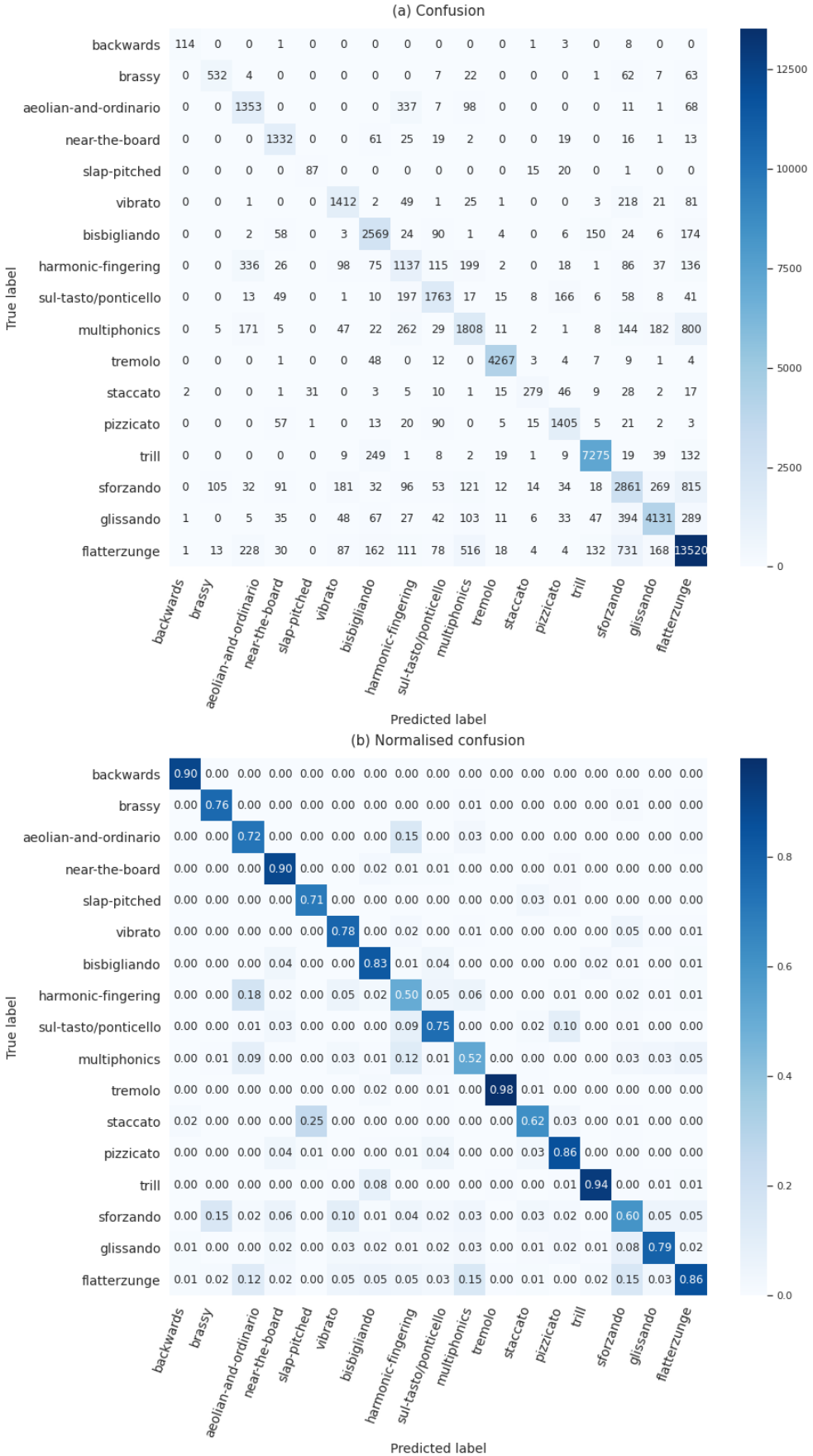

Fig. 3. Confusion matrices obtained for multi-class classification of techniques in the SOL dataset. (a): confusion matrix with number of frames detected; (b) normalised confusion matrix with (a) divided by the number of samples per technique.

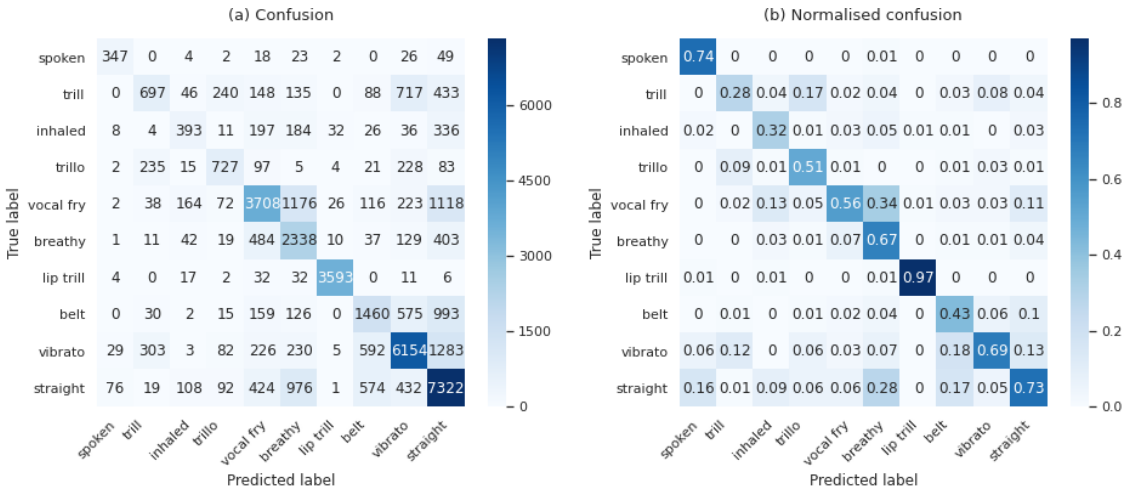

Fig. 4. Confusion matrices obtained for multi-class classification of ten vocal techniques. (a): confusion matrix with number of frames detected; (b) normalised confusion matrix with (a) divided by the number of samples per technique.

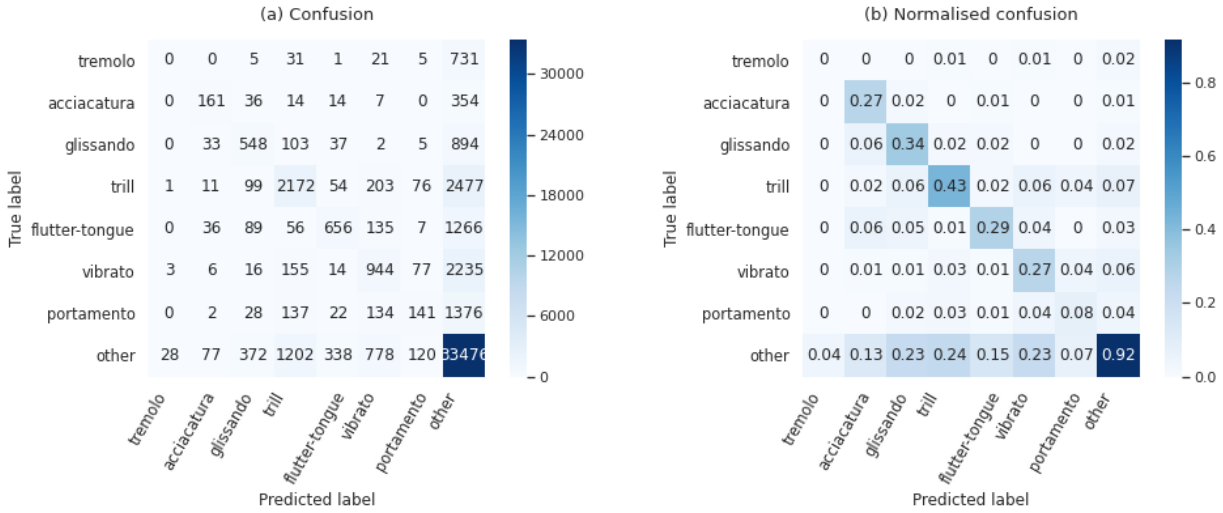

Fig. 5. Confusion matrices obtained for multi-class classification of playing techniques in the CBFdataset based on the MFCC features. (a): confusion matrix with number of frames detected; (b) normalised confusion matrix with (a) divided by the number of samples per technique.

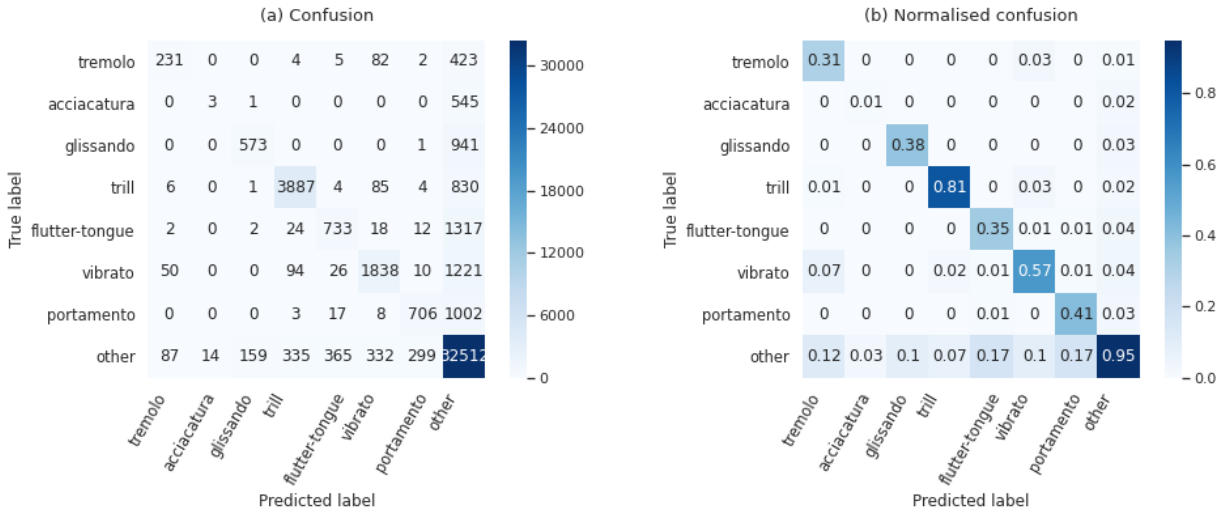

Fig. 6. Confusion matrices obtained for multi-class classification of playing techniques in the CBFdataset based on the MPS features. (a): confusion matrix with number of frames detected; (b) normalised confusion matrix with (a) divided by the number of samples per technique.

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